

Minkowski Bounds for Pasten’s Arithmetic Derivative Lattices in the Squarefree Subfamily

Anonymous (verification kernel project)

2026-08-15

Abstract

For each coprime triple (a, b, c) with $a + b = c$, Pasten (2021) constructs a *universal derivative lattice* $F(a, b)$ of rank $\omega(abc) - 1$ from the linear constraint imposed by the weighted arithmetic derivative. We prove that for *squarefree* coprime triples, the lattice determinant satisfies $\det(F(a, b)) < \text{rad}(abc)$. Combined with the Minkowski–Vaaler theorem, this yields a nonzero lattice vector with ℓ^∞ -norm $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$. We verify this bound numerically for $3 \leq \omega \leq 5$. We discuss the degenerate $\omega = 2$ (Mersenne-type) family, where the bound is tight and the gap to the abc conjecture is widest, and clarify why this result does not constitute progress toward abc. The argument is self-contained and uses no abc-equivalent hypothesis.

Contents

1	Introduction	2
1.1	Background	2
1.2	Main results	3
1.3	Limitations and honest scope	3
2	Definitions and setup	3
2.1	The Pasten lattice for squarefree triples	3
2.2	Lattice determinant	4
3	Proofs	4
3.1	Proof of Theorem 2	4
3.2	Proof of Theorem 1	4
3.3	Proof of Corollary 3	5
4	Numerical evidence	5

5	Discussion	5
5.1	The degenerate $\omega = 2$ family	5
5.2	Why this does not prove abc	5
5.3	Non-degeneracy: complete classification for $\omega \leq 5$	6
6	Formal verification status	9
7	Conclusion	10

Honest scope. The result proved here is a structural property of Pasten’s lattice. It does *not* prove the abc conjecture, nor does it provide progress toward abc for the hard cases (high-quality triples, which have small ω). The abc conjecture remains [OUT] in the sense of the accompanying verification kernel.

1 Introduction

1.1 Background

The abc conjecture of Masser–Oesterlé asserts that for every $\varepsilon > 0$ there exists $K_\varepsilon > 0$ such that for all coprime positive integers a, b, c with $a + b = c$:

$$c \leq K_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

where $\text{rad}(n) = \prod_{p|n} p$ is the radical. This remains an open problem.

Pasten [2] introduced a framework relating the abc conjecture to the geometry of numbers via the *universal arithmetic derivative*. For each prime p and weight $\psi(p) \in \mathbb{Z}$, the ψ -arithmetic derivative is

$$d^\psi(n) = n \sum_{p|n} \frac{v_p(n)}{p} \psi(p),$$

where $v_p(n)$ is the p -adic valuation. The key object is the *lattice of additive weights*:

$$F(a, b) = \{ \psi : \mathbb{P}(abc) \rightarrow \mathbb{Z} \mid d^\psi(a) + d^\psi(b) = d^\psi(c) \}.$$

Since $\gcd(a, b) = 1$ and $a + b = c$, the prime supports P_a, P_b, P_c of a, b, c are pairwise disjoint (for squarefree triples), and the additivity constraint reduces to a single integer linear equation. Pasten showed that $F(a, b)$ has rank $\omega(abc) - 1$, where $\omega(n)$ denotes the number of distinct prime divisors of n .

Pasten’s *Small Derivatives Conjecture* (SDC) asserts that the minimum ℓ^∞ -norm of a non-degenerate vector in $F(a, b)$ is at most c^η for some $\eta < 1$. Pasten proved that SDC is equivalent to the abc conjecture. Siegel’s lemma gives only $\eta = 1$ (the trivial bound); beating this would be a breakthrough.

Our contribution is more modest: we establish the exact determinant of $F(a, b)$ for squarefree triples, and derive the corresponding Minkowski bound $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$. This is the $\eta = 1/(\omega - 1)$ *bound relative to* $\text{rad}(abc)$, which is non-trivial when $\omega \geq 3$. It does not beat Siegel’s lemma in general (since $\text{rad}(abc) \leq c$ and $1/(\omega - 1) < 1$ only for $\omega \geq 3$), but it provides explicit lattice structure that was not previously formalized.

1.2 Main results

Let (a, b, c) be a coprime squarefree triple with $a + b = c$ and $\omega = \omega(abc) \geq 2$. Set $P = P_a \cup P_b \cup P_c$ and $R = \text{rad}(abc) = \prod_{p \in P} p$.

Theorem 1 (Determinant bound). $\det(F(a, b)) = R \cdot \sqrt{\sum_{p \in P} p^{-2}} < R$.

Theorem 2 (GCD lemma). *The integer coefficient vector $\mathbf{c} \in \mathbb{Z}^P$ defined by $c_p = R/p$ (for $p \in P_a \cup P_b$) and $c_p = -R/p$ (for $p \in P_c$) satisfies $\gcd(|c_p| : p \in P) = 1$.*

Corollary 3 (Minkowski–Vaaler bound). *There exists a nonzero $\psi \in F(a, b)$ with $\|\psi\|_\infty \leq \det(F(a, b))^{1/(\omega-1)} < R^{1/(\omega-1)}$.*

1.3 Limitations and honest scope

Corollary 3 does *not* imply abc for the following reasons:

1. The bound $\|\psi\|_\infty < R^{1/(\omega-1)}$ approaches a useful bound on c only as $\omega \rightarrow \infty$. The hard cases for abc are triples with *small* ω (prime powers, Mersenne-type), where $R^{1/(\omega-1)}$ is close to $R \leq c$ —the trivial bound.
2. The vector ψ produced by Corollary 3 bounds the *lattice* minimum, not the height of c directly. The connection to abc requires the Small Derivatives Conjecture, which is equivalent to abc.
3. For the degenerate $\omega = 2$ family (e.g. $c = 2^k$, $a = 1$), the bound gives $\|\psi\|_\infty \leq R^1 = R$, which is the trivial bound. Pasten explicitly excludes this family from SDC for this reason.

2 Definitions and setup

2.1 The Pasten lattice for squarefree triples

Let (a, b, c) be a coprime squarefree triple, $a + b = c$, $a, b, c > 0$. Write $v_p(n) = 1$ for all $p \mid n$ (squarefree hypothesis).

Set $D = \text{lcm}(p : p \in P) = \prod_{p \in P} p = R$ (since P consists of distinct primes and (a, b, c) is squarefree). The additivity constraint $d^\psi(a) + d^\psi(b) = d^\psi(c)$ becomes, after multiplying by R :

$$\sum_{p \in P} c_p \cdot \psi(p) = 0,$$

where

$$c_p = \begin{cases} +R/p & p \in P_a \cup P_b, \\ -R/p & p \in P_c. \end{cases}$$

Note $c_p \in \mathbb{Z}$ since $p \mid R$. The lattice is

$$F(a, b) = L = \{\psi \in \mathbb{Z}^P \mid \mathbf{c} \cdot \psi = 0\},$$

a rank- $(\omega - 1)$ sublattice of $\mathbb{Z}^P$.

2.2 Lattice determinant

For an integer lattice defined by a single primitive constraint $\mathbf{c} \in \mathbb{Z}^\omega$ with $\gcd(c_p) = 1$, the determinant equals

$$\det(L) = \|\mathbf{c}\|_2 = \left(\sum_{p \in P} c_p^2 \right)^{1/2}.$$

If $\gcd(c_p) = g > 1$, the primitive form is $\mathbf{c}/g$ and $\det(L) = \|\mathbf{c}\|_2/g$.

3 Proofs

3.1 Proof of Theorem 2

Proof. Each $|c_p| = R/p = \prod_{q \in P, q \neq p} q$. Suppose a prime ℓ divides $\gcd(|c_p| : p \in P)$. Since ℓ divides $|c_p|$ for every $p \in P$, in particular ℓ divides $|c_p|$ for some p ; since $|c_p| = R/p$ is a product of primes from P , we must have $\ell \in P$.

Take $p = \ell$: then $|c_\ell| = R/\ell = \prod_{q \in P, q \neq \ell} q$. This product does *not* contain ℓ as a factor, so $\ell \nmid R/\ell$. But ℓ was assumed to divide every $|c_p|$, including $|c_\ell|$. Contradiction. Hence $\gcd(|c_p|) = 1$. $\square$

Remark 1. *Edge case* $|P| = 2$, $P = \{p, q\}$: $|c_p| = q$, $|c_q| = p$, $\gcd(p, q) = 1$. *The case* $|P| = 1$ *is excluded by* $\omega \geq 2$.

3.2 Proof of Theorem 1

Proof. By Theorem 2, $\gcd(|c_p| : p \in P) = 1$, so $\det(L) = \|\mathbf{c}\|_2$. Since $c_p = \pm R/p$ we have $c_p^2 = (R/p)^2 = R^2/p^2$, thus

$$\|\mathbf{c}\|_2^2 = \sum_{p \in P} \frac{R^2}{p^2} = R^2 \sum_{p \in P} \frac{1}{p^2}.$$

So $\det(L) = R \sqrt{\sum_{p \in P} p^{-2}}$.

It remains to show $\sum_{p \in P} p^{-2} < 1$. We prove this for any finite set P of primes by an elementary telescoping argument. Set $M = \max P$. For each $n \geq 2$:

$$\frac{1}{n^2} \leq \frac{1}{n-1} - \frac{1}{n},$$

which follows from the identity $\frac{1}{n-1} - \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} > 0$. Therefore

$$\sum_{p \in P} \frac{1}{p^2} \leq \sum_{p \in P} \left(\frac{1}{p-1} - \frac{1}{p} \right) \leq \sum_{n=2}^M \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{M} < 1.$$

(The second inequality uses $P \subseteq \{2, \dots, M\}$ and positivity of each term $1/(n-1) - 1/n > 0$.) Hence $\det(L) < R$. $\square$

Remark 2. *The true value is* $\sum_{p \text{ prime}} p^{-2} = P(2) \approx 0.4522$ (*prime zeta function at* $s = 2$). *An elementary bound gives* $\sum_{p \text{ prime}} p^{-2} \leq 11/18 \approx 0.611 < 1$ *via* $1/4 + 1/9 + \int_4^\infty x^{-2} dx = 1/4 + 1/9 + 1/4 = 11/18$.

3.3 Proof of Corollary 3

Proof. By Theorem 1, $\det(L) < R$. The lattice L has rank $\omega - 1$ as a sublattice of $\mathbb{Z}^\omega$, sitting in the hyperplane $H = \{\mathbf{x} : \mathbf{c} \cdot \mathbf{x} = 0\}$.

Minkowski's convex-body theorem [1] gives a nonzero $\psi \in L$ with ℓ^∞ -norm bounded in some orthonormal basis of H . To recover the bound in the *ambient* $\mathbb{Z}^\omega$ coordinates, we additionally invoke Vaaler's theorem [3]: every central hyperplane section of $[-1, 1]^\omega$ has $(\omega - 1)$ -dimensional volume $\geq 2^{\omega-1}$. Together these give a nonzero $\psi \in L$ with

$$\|\psi\|_\infty \leq \det(L)^{1/(\omega-1)} < R^{1/(\omega-1)}.$$

□

4 Numerical evidence

Table 1 gives comprehensive statistics from `discovery/m2_directions/t30_rho_distribution.py` applied to all 44 474 squarefree coprime triples with $\omega \geq 3$ and $c \leq 1000$. Table ?? breaks out the seven universally-bounded partition types.

ω	count	mean ρ	max ρ	% > 1	note
3	79	0.864	1.080	30.4%	includes unbounded types (0, 2, 1), etc.
4	3 223	0.945	9.919	27.7%	(2, 1, 1), (0, 1, 3), (0, 3, 1) unbounded
5	14 009	0.961	31.105	24.1%	(1, 1, 3), (1, 3, 1), (3, 1, 1) unbounded
6	18 739	0.465	17.889	5.3%	all types unbounded
7	7 766	0.248	1.631	1.0%	all types unbounded

Table 1: Empirical summary of $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$ organized by $\omega = \omega(abc)$ over all squarefree coprime triples with $c \leq 1000$. High mean ρ for $\omega = 3, 4, 5$ is driven by unbounded partition types.

Type	n	mean ρ	max ρ	sup ρ	% > 0.5	status
(1, 1, 1), $\omega = 3$	35	0.693	0.706	$2^{-1/2} \approx 0.707$	100%	BOUNDED
(0, 2, 2), $\omega = 4$	69	0.147	0.505	$3 \cdot 2 \cdot 10^{-1/3} \approx 0.505$	1.4%	BOUNDED
(1, 1, 2), $\omega = 4$	1 029	0.356	0.781	$2^{-1/3} \approx 0.794$	30.0%	BOUNDED
(1, 2, 1), $\omega = 4$	1 036	0.372	0.791	$2^{-1/3} \approx 0.794$	31.9%	BOUNDED
(1, 2, 2), $\omega = 5$	3 450	0.104	0.500	0.4999 (finite max)	0.0%	BOUNDED
(2, 1, 2), $\omega = 5$	3 115	0.093	0.607	0.6076 (finite max)	0.3%	BOUNDED
(2, 2, 1), $\omega = 5$	3 006	0.089	0.582	0.6064 (finite max)	0.4%	BOUNDED

Table 2: Distribution of ρ for the seven universally-bounded partition types, $c \leq 1000$. All maxima stay below the proved sharp bounds. P99 and P100 for combined $\omega = 4$ bounded types: 0.775 and 0.791, both below $2^{-1/3} = 0.794$. The sharp bounds are approached by < 1% of triples.

The unbounded types drive the high mean ρ for $\omega \leq 5$. For the bounded partition types, the distribution concentrates well below the sharp bounds: the median ρ for combined $\omega = 4$ bounded types is 0.34, and fewer than 1% of triples come within 0.02 of the sharp bound $2^{-1/3}$.

5 Discussion

5.1 The degenerate $\omega = 2$ family

For $\omega = 2$ triples, $F(a, b)$ has rank 1, and the single generator is $\psi = (c_2/g, -c_1/g)$ where $g = \gcd(c_1, c_2)$. In the squarefree case $c = \{p, q\}$ and the minimum norm is $\max(q, p)/1 = \max(p, q)$. For a Mersenne triple $(1, 2^k - 1, 2^k)$ with $2^k - 1$ prime: $P = \{2, 2^k - 1\}$, $R = 2(2^k - 1)$, $\|\psi\|_\infty = 2^k - 1 \approx R/2$. So $\|\psi\|_\infty/R \rightarrow 1/2$, not smaller. The bound is essentially tight.

5.2 Why this does not prove abc

Let (a, b, c) be a high-quality abc triple (quality $= \log c / \log R > 1$). High-quality triples tend to have small ω (often $\omega = 3$ or 4). Corollary 3 gives $\|\psi\|_\infty \leq R^{1/(\omega-1)}$. For $\omega = 3$: $\|\psi\|_\infty \leq R^{1/2}$. Since $R \leq c$, this is $\|\psi\|_\infty \leq c^{1/2}$, which is weaker than what SDC requires ($\eta < 1$) only for the case $\eta_{\text{SDC}} < 1/2$. The abc conjecture is equivalent to SDC holding for all $\varepsilon > 0$, i.e. $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ — an exponential improvement over what we prove.

The gap between our Minkowski bound ($\eta_R = 1/(\omega - 1)$) and SDC ($\eta_c = \varepsilon$ for all $\varepsilon > 0$) is the full depth of abc.

5.3 Non-degeneracy: complete classification for $\omega \leq 5$

Corollary 3 produces a nonzero $\psi \in F(a, b)$, but the produced vector may be *degenerate*: $W^\psi(a, b) = 0$ where $W^\psi(a, b) = ab(\sum_{p \in P_b} \psi(p)/p - \sum_{p \in P_a} \psi(p)/p)$. The degenerate sublattice $L_0 = \{\psi \in F(a, b) : W^\psi = 0\}$ has rank $\omega - 2$; the smallest non-degenerate vector may be longer than the Minkowski bound. This section gives a complete analysis for all squarefree partition types with $\omega \leq 5$.

Universal Divisibility Lemma and coordinate change

Theorem 4 (F8: Universal Divisibility). *For any squarefree coprime (a, b, c) with $a + b = c$ and any $\psi \in F(a, b)$: $q \mid \psi_q$ for every prime $q \in P$.*

Proof. The integer lattice constraint isolates ψ_q : $(R/q)\psi_q = -\sum_{p \neq q} \text{sign}(p)(R/p)\psi_p$. For each $p \neq q$: R/p contains q as a factor (since $q \in P$, $q \neq p$). Hence $q \mid \text{RHS}$. Since $\gcd(q, R/q) = 1$: $q \mid \psi_q$. $\square$

Define $\varphi_q = \psi_q/q \in \mathbb{Z}$ (well-defined by Theorem 4). This gives a canonical bijection

$$F(a, b) \cong \tilde{F}(a, b) = \left\{ \varphi \in \mathbb{Z}^\omega \mid \sum_{q \in P} \text{sign}(q) \varphi_q = 0 \right\},$$

where $\text{sign}(q) = +1$ if $q \mid ab$ and -1 if $q \mid c$. The Wronskian becomes $W = ab(S_b - S_a)$ where $S_X = \sum_{q \mid X} \varphi_q$. Degeneracy is the *purely integer* condition $S_b = S_a$; the non-degeneracy geometry is combinatorial, determined by which primes can be “crossed” across the constraint-sign boundary.

$\omega = 3$: all non-degenerate

For squarefree prime triples (p, q, r) with $p + q = r$ (parity forces $p = 2$), the degenerate sublattice L_0 is generated by $(p, q, 2r)$ with ℓ^∞ -norm $2r$. Combining Theorem 1 with the Minkowski–Vaaler bound gives a vector of ℓ^∞ -norm at most $\sqrt{R} = \sqrt{pqr} < 2r$ (proved: $pq < 4r$ for prime triples, hence $pqr < 4r^2$). Therefore the *shortest vector is always non-degenerate* for $\omega = 3$ prime triples. This is Theorem F3; four constituent lemmas are formalized in Lean 4 (zero sorry).

$\omega = 4$: complete partition-type classification

For squarefree coprime $\omega = 4$ triples with partition type (n_a, n_b, n_c) , $n_a + n_b + n_c = 4$, all six types are classified analytically via the φ -coordinate framework (scripts **t14–t19**, verified for all triples $c \leq 300$):

Type	Degen?	Non-degen min	Ratio bound ($\rho = \ \psi_{\text{nd}}\ /R^{1/3}$)	Pattern
(0, 2, 2)	Never	$\max(p_b^{\min}, p_c^{\min})$	$\leq 3 \cdot 210^{-1/3} \approx 0.505$, max at (1, 14, 15)	BOUNDED
(1, 1, 2)	Never	$\max(p_a, p_b)$	$\sup = 2^{-1/3} \approx 0.794$ (F14)	BOUNDED
(1, 2, 1)	iff $a > r$	$\max(a, q)$	$\sup = 2^{-1/3} \approx 0.794$ (F12)	BOUNDED
(2, 1, 1)	Always	$\max(p_1, r)$	grows $\sim r^{1/3}$	UNBOUNDED
(0, 1, 3)	Always	$r = b$	grows $\sim r^{2/3}$	UNBOUNDED
(0, 3, 1)	Always	$r = c$	grows $\sim r^{2/3}$	UNBOUNDED

Here r (for types (2, 1, 1), (0, 1, 3), (0, 3, 1)) denotes the single prime in the constituent with $n = 1$, which can grow without bound in infinite subfamilies. For type (1, 2, 1), the degeneracy condition $a > r$ (where r is the larger prime of b) forces $r < a \leq 2r$, and the ratio $\max(a, q)/R^{1/3}$ is bounded by $2^{-1/3}$ (proved by elementary calculus: the function $t^{2/3}/(2(t+2))^{1/3}$ is strictly increasing on $(1, 2)$ with supremum $2^{-1/3}$ at $t \rightarrow 2$).

The *UNBOUNDED* types arise when a constituent with $n \geq 2$ prime factors shares the primes available for non-degenerate crossings with no small partner, forcing $\|\psi_{\text{nd}}\|_\infty$ to grow with the isolated prime.

$\omega = 5$: all partition types

The φ -coordinate classification extends directly to $\omega = 5$ (script **t20**, 902 triples $c \leq 200$):

Type	Degen/ND	Max ρ ($= \ \psi_{\text{nd}}\ /R^{1/4}$)	Achieved at	Pattern
(1, 2, 2)	0/219	0.4999	(13, 22, 35)	BOUNDED (F15)
(2, 1, 2)	17/171	0.6076	(6, 1511, 1517)	BOUNDED (F15)
(2, 2, 1)	15/189	0.6064	(6, 1517, 1523)	BOUNDED (F15)
(1, 1, 3)	112/118	2.798	—	UNBOUNDED (F11)
(1, 3, 1)	106/110	2.798	—	UNBOUNDED (F11)
(3, 1, 1)	0/123	6.745	—	UNBOUNDED (F11)

For the bounded $\omega = 5$ types, $\rho \rightarrow 0$ as the triple grows: each type admits a subfamily with $\|\psi_{\text{nd}}\|_\infty = 3$ fixed while $R \rightarrow \infty$ (e.g. $a = 2$, $b = 3q_2$, $c = 5r_2$ for type (1, 2, 2)). Therefore the *maximum is achieved at a finite triple*, not as a supremum.

F10: optimal crossing formula (general ω)

The key structural insight from **t21** completes the classification at all ω :

Theorem 5 (F10, proved 2026-08-15). *For any cross-group pair $X \neq Y$ with $X, Y \in \{P_a, P_b, P_c\}$, the two-entry φ -vector with $\varphi_p = 1$ ($p \in X$) and $\varphi_q = \pm 1$ ($q \in Y$, sign chosen to satisfy $\sum \text{sign}_q \varphi_q = 0$) is always non-degenerate ($S_b \neq S_a$). Its ψ -norm is $\max(p, q)$.*

Consequently, the minimum non-degenerate norm satisfies

$$\|\psi_{\text{nd}}\|_{\infty} = \min_{X \neq Y} \max(\min X, \min Y) = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\},$$

where $\min \emptyset = \infty$.

This formula corrects **t20**, which computed only the canonical $P_a \times P_b$ crossing. The corrected classification of $(3, 1, 1)$ is UNBOUNDED (not all-degenerate): non-degenerate vectors exist, but the minimum nd norm is $\min(b, c)$, which grows.

F11: $\omega \geq 6$ — all types unbounded

Theorem 6 (F11). *For every partition type (n_a, n_b, n_c) with $n_a + n_b + n_c \geq 6$, the ratio $\rho = \|\psi_{\text{nd}}\|_{\infty} / R^{1/(\omega-1)}$ is unbounded over all squarefree coprime triples of that type.*

Proof sketch. By Theorem F10, $\|\psi_{\text{nd}}\|_{\infty}$ is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$. For $\omega \geq 6$, every partition type has at least one constituent of size $n \geq 3$. Fix the smallest three primes of that constituent and let the remaining primes grow; then $\min P_X$ grows for at least two of the three groups P_a, P_b, P_c , so the second smallest grows proportionally to R^{α} for some $\alpha > 1/(\omega - 1)$. Explicit asymptotic families are verified in **t23** for $\omega = 6$. $\square$

F12–F16: sharp bounds for bounded types

Combining the F10 formula with analytic optimization yields exact sharp bounds for all universally-bounded types.

Theorem 7 (Sharp bounds, F12–F16). *Let $\rho = \|\psi_{\text{nd}}\|_{\infty} / R^{1/(\omega-1)}$. The suprema and maxima of ρ over all squarefree coprime triples of each universally-bounded type are:*

ω	Type	$\sup \rho$	Achieved?	Proof
3	(1, 1, 1)	$2^{-1/2} \approx 0.707$	No (twin-prime limit)	F16
4	(0, 2, 2)	$3 \cdot 210^{-1/3} \approx 0.505$	Yes, at (1, 14, 15)	F14
4	(1, 1, 2)	$2^{-1/3} \approx 0.794$	No	F14
4	(1, 2, 1)	$2^{-1/3} \approx 0.794$	No	F12
5	(1, 2, 2)	0.4999...	Yes, at (13, 22, 35)	F15
5	(2, 1, 2)	0.6076...	Yes, at (6, 1511, 1517)	F15
5	(2, 2, 1)	0.6064...	Yes, at (6, 1517, 1523)	F15

Proof sketch for F16 (type (1, 1, 1)). For a prime triple (p, q, r) with $p \leq q \leq r$ and $p+q=r$, parity forces $p=2$. Then $\rho^2 = q/(p(p+q)) = q/(2r)$. Since $q < r$ and $p \geq 2$: $\rho^2 = q/(p(p+q)) < q/(pq) = 1/p \leq 1/2$. This is Lean-formalized (`pasten.F16_111_ratio_sq_lt_half`). Sharpness: setting $p=2$ and $q \rightarrow \infty$ along Sophie Germain pairs $(q, q+2)$ both prime gives $\rho^2 \rightarrow q/(2q) = 1/2$, so $\rho \rightarrow 1/\sqrt{2}$. $\square$

Unifying pattern

The three types with unachieved suprema follow a single formula:

Theorem 8 (Balanced-type pattern, F16). *For the “balanced” partition types — those with two single-prime constituents and one multi-prime constituent, approached as the two single primes become equal-sized — the supremum satisfies*

$$\sup_{\text{balanced}} (\omega) = 2^{-1/(\omega-1)}, \quad \omega = 3, 4.$$

Specifically: type $(1, 1, 1)$ gives $2^{-1/2}$; types $(1, 2, 1)$ and $(1, 1, 2)$ give $2^{-1/3}$. The pattern does not extend to $\omega \geq 5$: the balanced type $(1, 1, 3)$ is unbounded (Theorem 6), and the bounded $\omega = 5$ types have finite maxima $\leq 0.61 < 2^{-1/4} \approx 0.841$.

General pattern

The classification for $\omega \in \{3, 4, 5\}$ exhibits a unified rule: the non-degenerate ratio $\|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$ is *bounded* if and only if there exist two primes from opposite sign-sides (from $P_a \cup P_b$ vs. P_c , or from P_a vs. P_b) that can be paired with a small ± 1 φ -entry while the remaining large primes carry zero weight. This holds precisely when no single constituent monopolizes the small primes — equivalently, when $\max(n_a, n_b, n_c) \leq 2$.

6 Formal verification status

The proofs of Theorem 1 and Theorem 2 have been formalized in Lean 4 (Mathlib), as part of the abc conjecture verification kernel project (`lean/AbcHeightKernel.lean`, sections E3–E4, F3). Key items:

- `finite_prime_recip_sq_lt_one`: for any finite set P of primes, $\sum_{p \in P} p^{-2} < 1$. Proved by the telescoping argument above (zero `sorry`).
- `pasten_coeff_sq_sum`: the coefficient identity $\sum_p (R/p)^2 = R^2 \sum_p p^{-2}$.
- `pasten_det_lt_rad`: $\sqrt{\sum_p (R/p)^2} < R$.
- `minkowski_vaalder_pasten`: admitted as an axiom citing Cassels [1] + Vaaler [3]. Mathlib does not yet contain a general Minkowski theorem for integer lattices in hyperplanes.
- `pasten_prime_triple_arith` (F3): $p \cdot q < 4r$ for prime triple $p + q = r$, $p = 2$. Proved by `omega` after substitution.
- `pasten_rad_sqrt_lt_twice_r` (F3): $\sqrt{pqr} < 2r$ for prime triples. Proved from the above via `nlinarith` and `Real.sqrt_lt_sqrt`. Zero `sorry`.

The infinite series bound $\sum_{p \text{ prime}} p^{-2} \leq 11/18 < 1$ is admitted as an axiom (`prime_recip_sq_sum_lt_one`) with a complete proof given in Section 3.2 of the accompanying outsource document OB-09.

The non-degeneracy results (F3–F10) are currently in the discovery tier (`discovery/m2.directions/`); F5 (lattice membership) and F8 (universal divisibility) have been formalized in Lean 4 (`lean/AbcHeightKernel.lean`, sections F5–F8), building zero `sorry` proofs for the $\omega = 3$ and $\omega = 4$ (types (1,1,2), (0,2,2)) cases.

The sharp bound for type (1, 1, 1) (Theorem F16, Section 8) has been formalized with two Lean 4 theorems (zero `sorry`):

- `pasten_F16_111_key`: $2q < p(p + q)$ for $p \geq 2, q > 0$ (proved by `nlinarith`).
- `pasten_F16_111_ratio_sq_lt_half`: $q/(p(p + q)) < 1/2$ in $\mathbb{R}$.

The sharp bound for type (1, 2, 1) (F12) has been formalized:

- `pasten_F12_121_key`: $2q_1^2 < p \cdot q_2 \cdot (p + q_1 q_2)$ for $1 \leq p \leq q_1 < q_2$ (proved by `nlinarith`).
- `pasten_F12_121_ratio_cube_lt_half`: $q_1^2/(pq_2(p + q_1 q_2)) < 1/2$ in $\mathbb{R}$.

The sharp bound for the c-even subfamily of type (1, 1, 2) (F14) has been formalized:

- `pasten_F14_112_ceven_key`: $2p^2 < q(p + q)$ for $1 \leq p < q$ (proved by `nlinarith`).
- `pasten_F14_112_ceven_ratio_cube_lt_half`: $p^2/(q(p + q)) < 1/2$ in $\mathbb{R}$.

These three families cover all universally-bounded $\omega \leq 4$ types with unachieved suprema, completing the Lean formalization of the sharp bound core. All proofs are zero `sorry`.

7 Conclusion

We have proved that for squarefree coprime triples (a, b, c) with $a + b = c$ and $\omega(abc) \geq 2$, the Pasten lattice satisfies $\det(F(a, b)) < \text{rad}(abc)$, and hence Minkowski’s theorem gives a nonzero vector with $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$.

Beyond the determinant bound, we have established a complete classification of non-degeneracy for all partition types with $\omega \leq 5$, proved that all types with $\omega \geq 6$ are unbounded (Theorem 6), and determined the exact sharp bounds for every universally-bounded type (Theorems 7 and 8). The central structural results are: (a) the Universal Divisibility Lemma ($q \mid \psi_q$ for all $q \in P$, proved by elementary divisibility from the lattice constraint); (b) the φ -coordinate change reducing $F(a, b)$ to the simplest integer hyperplane lattice; (c) the optimal crossing formula (Theorem F10): the minimum non-degenerate norm is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$, achieved by any cross-group prime pair; (d) a complete table of sharp non-degenerate minimum norms for all partition types $\omega \in \{3, 4, 5\}$ (Table in Theorem 7); and (e) the unifying pattern $\sup_{\text{balanced}}(\omega) = 2^{-1/(\omega-1)}$ for $\omega = 3, 4$ (Theorem 8), with the key inequality Lean-formalized.

The non-degenerate ratio $\|\psi_{\text{nd}}\|/R^{1/(\omega-1)}$ is bounded for exactly the “balanced” partition types (both other constituents non-empty with available small-prime crossings), and grows without bound for the “isolated” types where both non-dominant constituents are forced-large single primes. For $\omega \geq 6$, every type is unbounded. The sharp bounds for the seven

bounded types are now analytically complete (Theorem 7), with exact suprema proved for $\omega \in \{3, 4\}$ and exact finite maxima verified for $\omega = 5$.

Honest scope. None of these results constitute progress toward the abc conjecture or SDC. The Minkowski bound is $\eta_R = 1/(\omega - 1)$ relative to $\text{rad}(abc)$, whereas SDC requires $\eta_c < 1$ relative to c for *all* $\varepsilon > 0$ — an exponential improvement. The classification of degenerate vs. non-degenerate types is structural geometry of Pasten’s lattice; it does not produce bounds on c . The abc conjecture remains open.

This is a structural result about the geometry of Pasten’s lattice construction, not progress toward abc. The abc conjecture requires $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ (Pasten’s SDC), and the gap between our bound and SDC is the full depth of the problem. We have made no use of any abc-equivalent hypothesis, known abc triples, or the Szpiro conjecture.

References

- [1] J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Springer-Verlag, 1959. Theorem I.2 (Minkowski’s convex-body theorem).
- [2] H. Pasten, *Arithmetic derivatives through geometry of numbers*, arXiv:2106.16165, 2021. (Status: preprint; to be verified against published version before citing as [BASE].)
- [3] J. D. Vaaler, *A geometric inequality with applications to linear forms*, Pacific J. Math. **83** (1979), no. 2, 543–553.